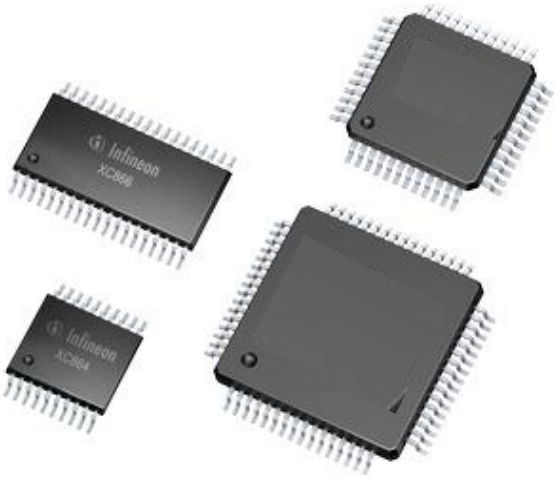
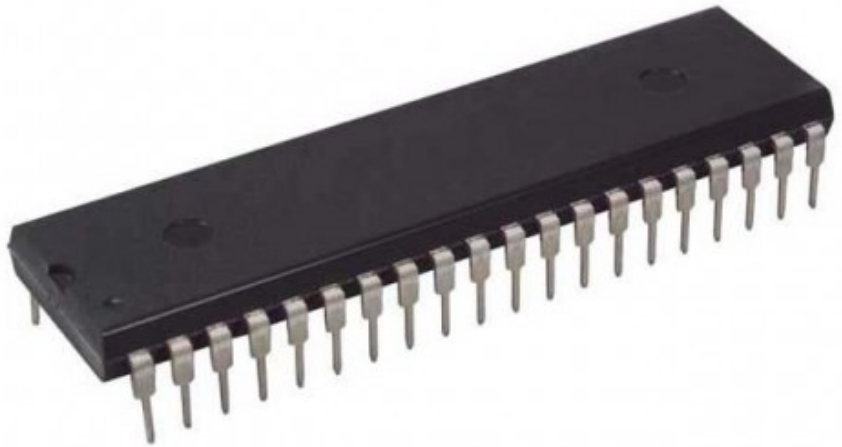
8051 Microcontroller





Delay calculation

```
        ORG 00h

HERE:   MOV P0,#0FFH
        LCALL DELAY
        MOV P0,#00H
        LCALL DELAY
        SJMP HERE

DELAY:  MOV R0,#m

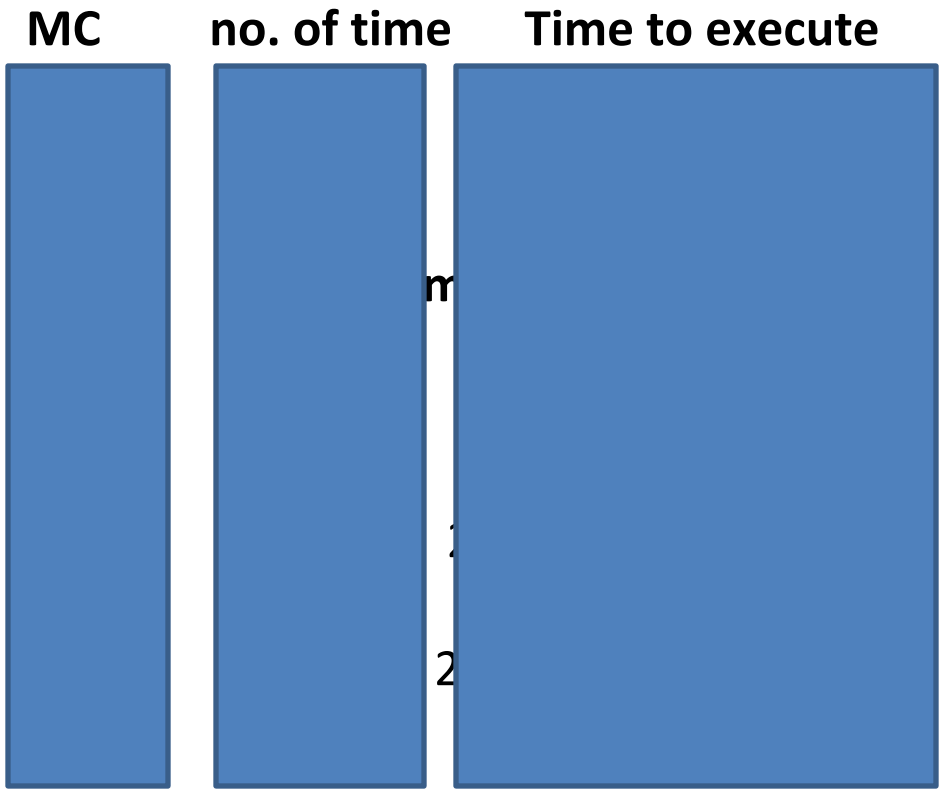
HERE2:  MOV R1,#n

HERE1:  DJNZ R1,HERE1

        DJNZ R0,HERE2

        RET

        END
```



Delay calculation

```
        ORG 00h

HERE:   MOV P0,#0FFH
        LCALL DELAY
        MOV P0,#00H
        LCALL DELAY
        SJMP HERE

DELAY:  MOV R0,#180

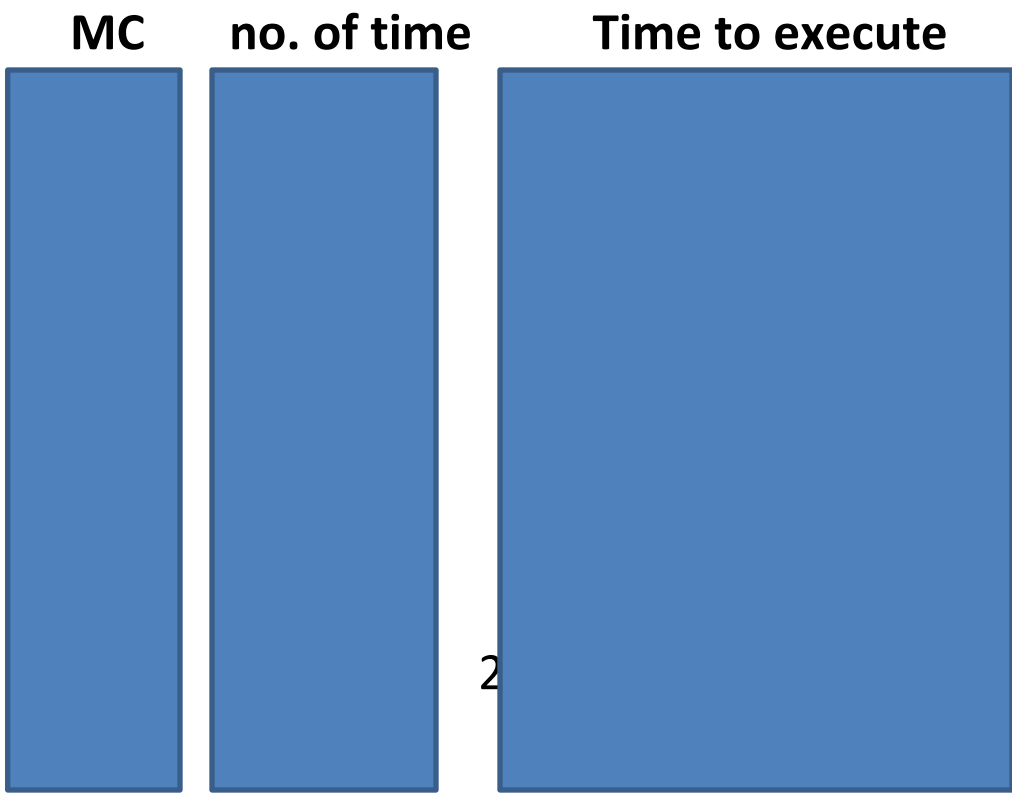
HERE2:  MOV R1,#255

HERE1:  DJNZ R1,HERE1

        DJNZ R0,HERE2

        RET

        END
```



Delay calculation

```
        ORG 00h

HERE:   MOV P0,#0FFH
        LCALL DELAY
        MOV P0,#00H
        LCALL DELAY
        SJMP HERE

DELAY:  MOV R0,#180

HERE2:  MOV R1,#255

HERE1:  DJNZ R1,HERE1

        DJNZ R0,HERE2

        RET

        END
```

MC	no. of time	Time to execute
1MC	1	1
1MC	**180**	**180**
2MC	**255** x **180**	2 x **255** x **180** = 91800
2MC	**180**	2 x **180** = 360
2MC	1	2

Delay calculation

```
        ORG 00h

HERE:   MOV P0,#0FFH
        LCALL DELAY
        MOV P0,#00H
        LCALL DELAY
        SJMP HERE

DELAY:  MOV R0,#n
HERE3:  MOV R1,#255
HERE2:  MOV R2,#255
HERE1:  DJNZ R2,HERE1
        DJNZ R1,HERE2
        DJNZ R0,HERE3
        RET

        END
```

2 x **255** x **255** x **n** x **1.085us** = **1** sec